

Documentation

Data Cleaning Logic

- Imported data using **Power Query “From Folder”** to ensure automation and scalability.
 - Removed or corrected invalid records where timestamp sequences were illogical (e.g., gate-out before gate-in).
 - Handled missing values:
 - Single missing timestamps were estimated using average stage durations.
 - Rows with multiple missing timestamps were excluded to maintain data integrity.
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2. Assumptions Made

- Shift timings defined as:
 - Morning: 6 AM – 2 PM
 - Evening: 2 PM – 10 PM
 - Night: 10 PM – 6 AM
 - Missing timestamps were approximated using average durations of the respective stages.
 - Travel distance between PCT and SCY assumed constant (3.5 km), with no route variation.
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3. Final Operational Recommendations

- Improve SCY Yard efficiency through better resource allocation and scheduling.
- Deploy real-time monitoring and queue management systems to reduce delays and improve coordination.